



**Bournemouth
University**

Submission Coursework Brief

School of Computing and Engineering

Unit: Technological Innovations in Computing (COMP5067)		
Unit Contact: Gernot Liebchen	Credits: 20	Level: 5

Assessment Title: AR-Enhanced Maintenance Support System for High-Security Engineering Environments	
Assessment Number: 1 of 2	
Assessment Type: Group	Weighting (%): 70%
Deadline: 12/05/2026 12:00 PM	Submission Method: Brightspace assignment (Turnitin) / Brightspace assignment (Large/multiple file)

Quality Assessor (QA): Lai Xu	Other Marker(s): Festus Adedoyin Fudong Li Tim Orman
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Can I use Generative AI tools?

Basic spelling and grammar correction tools are permitted.

Generative AI task completion

Generative AI can be used to complete specified elements of the coursework as detailed below. Any generative AI content must be acknowledged. Any elements not specified must be completed without the use of generative AI.

This is a group assignment.

Students will work in a team of 4-5 members. Team members need to come from a mixture of pathways, such as:

- **3 Computing pathway students and 2 Cyber Security pathway students**
- **3 Computing pathway students and 2 Data Analytics pathway students**
- **2 Computing pathway students and 2 Cyber Security pathway students**

- **2 Computing pathway students and 2 Data Analytics pathway students**

Groups need to have at least 2 students of any pathway. There should be no group with pathways only represented by one student.

Students can choose their own group by completing the team signup form on Brightspace by 4pm 26th February 2026. Students will be randomly assigned to a group after 26th February 2026. Any concerns regarding group allocation must be raised with the Unit Leader by **4pm on 02/03/2026**.

Context: AR-Enhanced Maintenance Support System for Public Transport Environments

Public transport systems such as rail networks, metro stations, tram systems, and bus depots are complex, safety-critical environments that must operate reliably to ensure public safety and service continuity. These environments include both public-facing areas and restricted operational zones, where authorised engineers and technicians carry out inspection, maintenance, and repair activities.

Maintenance staff are required to identify, communicate, and resolve faults accurately and efficiently. These faults may include structural wear, material stress, electrical issues, signalling anomalies, or equipment degradation. Some faults may not be immediately visible to the naked eye, particularly in tunnels, service corridors, plant rooms, or behind installed infrastructure.

Public transport environments are also exposed to deliberate security threats. Threat actors may attempt to interfere with infrastructure, steal sensitive operational information, falsify maintenance data, or create physical safety risks for passengers and staff. As such, maintenance support systems must be designed not only for operational efficiency but also for **security, resilience, and trustworthiness**.

Augmented Reality (AR) technologies can enhance maintenance operations by enabling authorised personnel to visualise fault locations, overlay contextual information onto physical infrastructure, and collaborate securely in real time (see Figure 1). AR can also support **tool accountability**, ensuring that equipment used in maintenance tasks is tracked, verified, and auditable throughout an operation (see Figure 2).

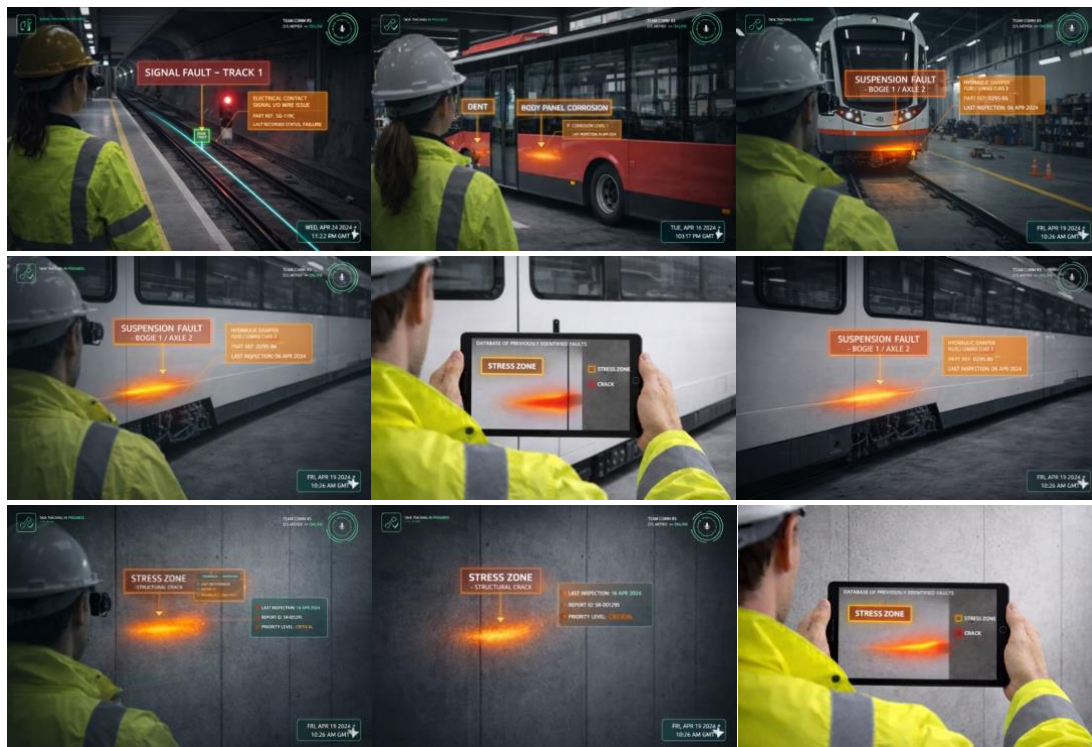


Figure 1: Top Row: Left – Example POV(Point of View) of AR System for London Underground. Middle – Example POV of AR System for Bournemouth Buses. Right – Example of POV of AR System for National Rail.
Middle Row: Left – Example POV of Train Side Inspection. Middle – Example POV Train Inspection AR on Tablet. Right – Example Screen for Train Inspection.
Bottom Row: Left – Example POV of Wall Inspection. Middle – Example POV Wall Inspection AR on Tablet. Right – Example Screen for Wall Inspection.



Figure 2: Left – Example of AR Tool Check. Right – Example AR Tool Check Mock-up.

Tasks:

You are required to design and develop a **prototype AR-based maintenance support system** for a **public transport context** outlined in Figure 3.

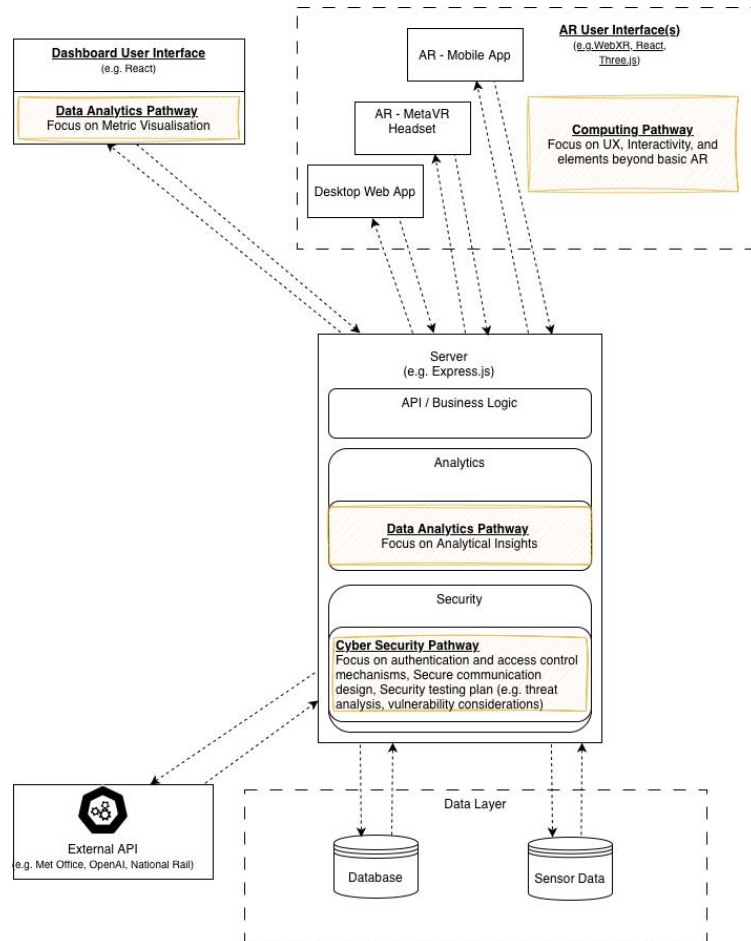


Figure 3: *Example Architecture of the Public Transport AR Prototype System (the yellow outlines indicative pathway specific focuses)*

The required prototype system should demonstrate the following:

1. Fault Detection and Data Visualisation with Backend Connection
 - Simulate faults using markers placed on walls, stations, tunnels, vehicles, or service areas (see Figure 1)
 - Use AR devices (mobile phones are sufficient) to detect, visualise, and annotate these faults
 - Enable secure sharing of fault information between authorised users
2. Tool Tracking and Accountability
 - Simulation of AR tool checks (see Figure 2)
 - AR-assisted recognition and tracking of tools during maintenance activities
 - Accurate recording and counting of tools to reduce safety and security risks
3. Security, Access Control, and Threat Awareness

- Authentication and access control for users and devices
 - Protection of system communications and stored data
 - Demonstrate how the system design reduces the risk of interference from threat actors, including actions that could compromise passenger safety, disrupt services, or expose sensitive infrastructure information.
4. System Integration and Monitoring Using Data Science
- Secure interaction between AR devices and a central dashboard
 - Provide monitoring features that support detection of unusual or suspicious system behaviour.

The prototype should demonstrate feasibility at **Technology Readiness Level 3 (TRL 3)** and reflect realistic constraints typical of high-security engineering environments. Your solution should integrate elements of software development, cybersecurity, and data handling, and analytics and predictive modelling, demonstrating how emerging technologies can enhance maintenance efficiency and reliability in public transport contexts.

Suggested Technologies and Architecture

The system architecture illustrated in Figure 3 represents a **suggested implementation approach** intended to support feasibility at **TRL 3**. Groups are encouraged to use **widely adopted, web-based technologies** suitable for AR, backend services, security, and data handling.

Indicative examples include:

- **Web-based AR** using standards such as **WebXR** (e.g. via the WebXR Device API)
- 3D rendering and interaction using frameworks such as **Three.js** or similar
- Web-based frontends implemented using frameworks such as **React** (e.g. for AR interfaces and dashboards)
- A backend server implemented using technologies such as **Node.js with Express.js** to support APIs, authentication, and secure communication
- Standard data storage (e.g. NoSQL, SQL, or Google Cloud) and visualisation technologies appropriate to the chosen approach

Groups may adapt or modify the suggested architecture and technology choices **where sufficient justification is provided** in the report. Alternative implementations are acceptable provided the system remains feasible, integrated, and appropriate to a proof-of-concept.

Role Expectations:

All Group Members

- Requirements analysis and design of the artefact, including functional and non-functional requirements
- Definition of user needs, assumptions, and the overall system structure

- Implementation of the core system infrastructure derived from the agreed design, including:
 - a basic AR user interface supporting fault and tool visualisation
 - backend services (including server and API)
 - a basic dashboard for data management
 - baseline security measures, such as authenticated access and protected data exchange
- System integration, testing, and evaluation of the overall system
- Final artefact and presentation

Computing Pathway Students

- Design and implement the AR user experience (UX), focusing on usability, clarity of overlays, and effective presentation of fault and tool information
- Develop interactive AR workflows (e.g. selection, annotation, confirmation, guided steps, feedback and error handling)
- Implement enhanced AR functionality beyond basic visualisation, such as persistent markers, multi-step inspection flows, or context-aware interactions
- Critical evaluation and justification of technical implementation decisions, including technical constraints, feasibility, and trade-offs encountered during development

Cyber Security Pathway Students

- Authentication and access control mechanisms
- Secure communication design
- Security testing plan (e.g. threat analysis, vulnerability considerations)
- Evaluation and justification of key security decisions, including technical constraints and trade-offs

Data Analytics Pathway Students

- Data flows and dashboard design
- Visualisation of faults, tools, and system metrics
- Analytical insights derived from system data
- Evaluation and justification of key data visualisation and algorithm decisions, including technical constraints and trade-offs

Each group must integrate all contributions into a single functioning prototype and a coherent written report.

Intended Learning Outcomes (ILOs)

1. Understand the theory and techniques involved in project management, and apply this knowledge to the planning, monitoring, control and evaluation of a project in a sustainable and ethical manner
2. Develop solutions based on the understanding of how the digital age works and the opportunities it presents by using existing programming frameworks and libraries.

3. Demonstrate the ability to work in a team in applying technological solutions to problem-solving.
4. Understand the range of new and emerging technologies in computing and how they influence business operations globally.

Submission Format:

There are four compulsory submissions for this assessment. All components must be submitted by the deadline to the Technological Innovations unit area on **Brightspace**.

1. Group Report

The group report must:

- Be submitted in PDF format.
- Maximum/expected is 15 pages (including figures, tables, references, and appendices).
- Include the final word count on the front page.

Please see the suggested report structure at the end of this brief.

2. Artefact

You must develop, based on the composition of your group (i.e. pathway membership), a **functional Proof of Concept prototype** aligned with the design documented in the report. The prototype should demonstrate **TRL 3** functionality.

The prototype must include the following elements:

- **All Groups (Core Prototype – mandatory)**

The following elements must be implemented by all groups, and all pathways are expected to contribute to the baseline system:

- basic AR interface supporting the identification and visualisation of simulated invisible faults
- AR-assisted tool tracking
- Baseline security measures, such as authenticated access and protected data exchange between system components

- **Computing Pathway Contribution**

The following elements must be designed and implemented by **Computing pathway students**:

- An enhanced AR UX that improves usability and clarity beyond the baseline AR interface
- Interactive AR features supporting meaningful user interaction (e.g. selection, annotation, confirmation, or guided steps)
- AR functionality beyond basic visualisation, appropriate to the project context
- Evaluation and justification of key implementation decisions, including technical constraints and trade-offs

- **Cyber Security Pathway Contribution**

The following element must be designed and implemented by **Cyber Security pathway students**:

- Authentication and access control mechanisms
- Secure communication
- Security testing plan
- Evaluation and justification of key security decisions, including technical constraints and trade-offs

- **Data Analytics Pathway Contribution**

The following elements must be designed and implemented by **Data Analytics pathway students**:

- A dashboard aggregating and visualising system data
- A machine learning algorithm demonstrating predictive modelling using system data
- Evaluation and justification of key data visualisation and algorithm decisions, including technical constraints and trade-offs

All elements must be integrated into a **single coherent prototype**, with clear evidence of collaboration between pathways.

The prototype should be **appropriate in scale and complexity** for demonstrating the feasibility of the proposed AR-based maintenance system. Fully deployed or production-ready solutions are **not expected**.

3. Group Presentation

Your presentation must:

- Be based on the group report and artefact.
- Include representative contributions from all group members.

Be submitted as:

- A PowerPoint file (maximum 15 slides)
- A video recording (MP4, at least 720p quality, maximum 15 minutes)

The presentation should clearly explain the concept, design rationale, prototype demonstration, and group reflections.

4. Individual Contribution Form

Each student must submit a completed Individual Contribution Form (PDF).

This form documents your contributions and is used to adjust marks where necessary. To resolve potential group contribution conflict, a workplace (e.g. a shared OneDrive folder) where the contribution of each group member can be tracked needs to be set up as soon as the project starts.

The deadline for all components, report, artefact, presentation slides, video, and contribution form, is the same.

How will this be assessed?

The Bournemouth University Generic Assessment Criteria for L5 will be applied.

The marks will be awarded as follows:

Criteria	Available Marks	Evidence Submitted	ILOs
Group Report • Project planning, scope, and Gantt chart • Requirements analysis and system design • Justification of technical decisions • Documentation of pathway roles and contributions • Evaluation, reflection, and academic quality	30%	Group report (PDF)	1, 2, 3, 4
Artefact • Core Integrated Prototype • Functional AR prototype (fault visualisation, tool tracking) • Backend connectivity and baseline security • Integration across AR, server, and dashboard components • Feasibility at TRL 3	20%	Demonstrated artefact	2, 3
Artefact – Pathway Contributions • <i>Computing pathway:</i> ○ AR user experience and interaction design ○ Enhanced AR functionality beyond basic visualization ○ Justification of technical and design decisions • <i>Cyber Security pathway:</i> ○ Authentication and access control mechanisms ○ Secure communication design ○ Security testing considerations ○ Justification of security decisions • <i>Data Analytics pathway:</i> ○ Dashboard design and data visualization ○ Descriptive or predictive data analysis ○ Use of analytics to support monitoring or decision-making ○ Justifications of data visualisation and algorithm decisions	20%	Demonstrated artefact + report linkage	2, 3, 4
Code Quality • Source Files & Setup Documentation • Organisation and clarity of source files • Quality of setup / deployment instructions • Reproducibility and ease of review	10%	Source code + setup instructions	1, 3
Group Presentation • Explanation of system concept and design rationale • Artefact demonstration aligned with report • Communication of pathway contributions • Overall clarity, structure, and professionalism	10%	Slides + recorded video	1, 2, 3
Innovation & Technical Integration	10%	Report + artefact + presentation	4

• Originality or creative integration of emerging technologies • Effective integration of AR, security, and data capabilities • Appropriateness to context and constraints			
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Questions about this assessment:

There will be in-class questions and answers about the brief, dedicated sessions during the labs, and office hours for answering questions about the brief

Academic Integrity

The work you submit must be your own. Any attempt to gain an unfair advantage in your assessment by **cheating**, deception or fraud is considered an academic offence. The 'Assessment help and support' section of the unit (found under 'Assessment' in the content area) provides more guidance on avoiding academic offences, including **any guidance on what will or will not be considered an academic offence in this specific assessment.**

Help and support

The 'Assessment help and support' section of the unit (found under 'Assessment' in the content area) provides information and guidance, including specific information on support for this assessment. It provides help with our policies on deadline extensions and information on support available in the university, including academic skills support and additional learning support for students with disabilities.

Disclaimer: The information provided in this assignment brief is correct at time of publication. In the unlikely event that any changes are deemed necessary, they will be communicated clearly via e-mail and via the VLE and a new version of this coursework brief will be circulated.

Version: 1

Updated: January 2026

Issued Date: 20/02/2026

Suggested Report Structure:

Introduction

- Contextualise **public transport maintenance challenges** and the role of Augmented Reality (AR) in supporting safety, efficiency, and reliability
- Outline the project goals, system concept, and chosen public transport context

Project Planning

- Project scope and objectives
- Work packages and pathway responsibilities
- Gantt chart
- Core system overview, including key components and interactions

Requirements Analysis and Design of the Artefact

- Functional and non-functional requirements for the AR-based maintenance system
- User needs (e.g. maintenance engineers, technicians, dashboard operators)
- Hardware and software assumptions
- AR interaction design (baseline behaviour)
- Object/fault detection logic
- Database design (ERD)
- UML diagrams (Use Case, Class, Activity ×2, Deployment or DevOps Pipeline)

Artefact Development

a) Core System Infrastructure (All Group Members)

- Implementation of the baseline system infrastructure derived from the agreed design
- Basic AR interface supporting fault and tool visualisation
- Backend services (including server and API)
- Basic dashboard for data management
- Baseline security measures (e.g. authenticated access, protected data exchange)

b) Enhanced AR User Experience and Interactivity (Computing Pathway)

- Enhanced AR user experience (UX) beyond the baseline interface
- Interactive AR features supporting meaningful user interaction
- AR functionality beyond basic visualisation, appropriate to the project context
- Evaluation and justification of key technical implementation decisions

c) Cyber Security Components (Cyber Pathway)

- Authentication and access control mechanisms
- Secure communication between AR devices, backend services, and dashboard
- Security testing plan (e.g. threat modelling, vulnerability assessment)
- Evaluation and justification of key security decisions

d) Data Architectures and Analytics (Data Pathway)

- Data flow diagrams for dashboards (fault logs, tool tracking, system metrics)
- Dashboard visualisation framework (e.g. Tableau, PowerBI, or equivalent)
- Predictive or descriptive analytics using system data

- [Evaluation and justification of key data visualisation and algorithm decisions](#)

e) Integrated Functional Prototype (All Group Members)

- Integrated AR prototype demonstrating:
 - detection of simulated invisible faults
 - AR-assisted tool tracking
 - secure multi-device interaction
 - dashboard synchronisation
- Evidence of testing and evaluation at **TRL 3**

5. Artefact Evaluation

- Evaluation of the prototype against requirements and design goals
- Reflection on system performance, limitations, and feasibility
- Consideration of security, usability, and data-handling effectiveness

6. Conclusion

References

- **APA style**

Appendices

- Additional diagrams, screenshots, testing logs, or supporting material